Derivatives of trascendental functions

In this session[24] we are going to delve into the derivatives of trascendental functions.

[24] February 5

Let's start with the derivative of the function $f(x) = e^x$.

$$
\begin{aligned}
\frac{\mathrm{d}e^x}{\mathrm{d}x} &= \lim_{h\to 0} \frac{e^{x+h} - e^x}{h} \\
&= \lim_{h\to 0} \frac{e^x e^h - e^x}{h} \\
&= \lim_{h\to 0} \frac{e^x\left(e^h - 1\right)}{h} \\
&= e^x \cancelto{1}{\lim_{h\to 0} \frac{\left(e^h - 1\right)}{h}} \\
&= e^x
\end{aligned}
$$

Using the numerical approach to compute $\lim_{h\to 0} \frac{\left(e^h-1\right)}{h}$:

h	$\frac{\left(e^h-1\right)}{h}$
0.01	1.005016708416795
0.001	1.0005001667083846
0.000001	1.0000004999621837

h	$\frac{\left(e^h-1\right)}{h}$
-0.01	0.9950166250831893
-0.001	0.9995001666249781
-0.000001	0.9999994999843054

$$\lim_{h\to 0} \frac{\left(e^h-1\right)}{h}$$

Now let's compute the derivative of $\ln[x]$:

$$
\begin{aligned}
\frac{\mathrm{d}\ln[x]}{\mathrm{d}x} &= \lim_{h\to 0} \frac{\ln[x+h] - \ln[x]}{h} \\
&= \lim_{h\to 0} \frac{1}{h} \ln\left[\frac{x+h}{x}\right] \\
&= \lim_{h\to 0} \frac{1}{h} \ln\left[1 + \frac{h}{x}\right] \\
&= \lim_{h\to 0} \ln\left[\left(1 + \frac{h}{x}\right)^{1/h}\right]
\end{aligned}
$$

Let's make the following change of variable $t = \frac{h}{x}$, therefore, $h = tx$. Computing the limit of t as $h \to 0$ to make the correct change of variable, $\lim_{h\to 0} \frac{h}{x} = 0$.

$$
\begin{aligned}
\frac{\mathrm{d}\ln[x]}{\mathrm{d}x} &= \lim_{t\to 0} \ln\left[(1+t)^{1/tx}\right] \\
&= \frac{1}{x} \cancelto{1}{\lim_{t\to 0} \ln\left[(1+t)^{1/t}\right]} \\
&= \frac{1}{x}
\end{aligned}
$$

Using the numerical approach to compute $\lim_{t\to 0} \ln\left[(1+t)^{1/t}\right]$:

h	$\ln\left[(1+t)^{1/t}\right]$
0.01	0.9950330853168091
0.001	0.9995003330834231
0.000001	0.9999994999180668

h	$\ln\left[(1+t)^{1/t}\right]$
-0.01	1.005033585350145
-0.001	1.0005003335835343
-0.000001	1.000000500029089

$$\lim_{t\to 0} \ln\left[(1+t)^{1/t}\right] = 1$$

Now it is time to compute the derivative of $\sin(x)$,

$$\frac{\mathrm{d}\sin(x)}{\mathrm{d}x} = \lim_{h\to 0} \frac{\sin(x+h) - \sin(x)}{h}$$

To avoid the $1/h$, let's manipulate the expression with the following trigonometric identity $\sin(x+h) - \sin(x) = 2\cos\left(x + \frac{h}{2}\right)\sin\left(\frac{h}{2}\right)$

$$
\begin{aligned}
\frac{\mathrm{d}\sin(x)}{\mathrm{d}x} &= \lim_{h\to 0} \frac{2\cos\left(x+\frac{h}{2}\right)\sin\left(\frac{h}{2}\right)}{h} \\
&= \left[\lim_{h\to 0} \cos\left(x + \frac{h}{2}\right)\right]\left[\lim_{h\to 0} \frac{2\sin\left(\frac{h}{2}\right)}{h}\right]
\end{aligned}
$$

Computing the limits separately,

$$
\begin{aligned}
\lim_{h\to 0} \cos\left(x + \frac{h}{2}\right) &= \cos\left(x + \frac{0}{2}\right) \\
&= \cos(x)
\end{aligned}
$$

For the next limit, we are going to make the following change of variable $u = \frac{h}{2}, \quad h = 2u$. Computing the limit to ensure the change of varaible we get that as $h \to 0$, $u \to 0$. Hence,

From previous excercises we already know that $\lim_{u\to 0} \frac{\sin(u)}{u} = 1$.

$$
\begin{aligned}
\lim_{h\to 0} \frac{2\sin\left(\frac{h}{2}\right)}{h} &= \lim_{u\to 0} \frac{2\sin(u)}{2u} \\
&= \cancelto{1}{\lim_{u\to 0} \frac{\sin(u)}{u}} \\
&= 1
\end{aligned}
$$

Replacing this results at the original limit,

$$
\begin{aligned}
\frac{\mathrm{d}\sin(x)}{\mathrm{d}x} &= \left[\lim_{h\to 0} \cos\left(x + \frac{h}{2}\right)\right]\left[\lim_{h\to 0} \frac{2\sin\left(\frac{h}{2}\right)}{h}\right] \\
&= [\cos(x)]\,[1] \\
&= \cos(x)
\end{aligned}
$$

To finish the session we are going to show the derivative of $\cos(x)$.

$$
\frac{\mathrm{d}\cos(x)}{\mathrm{d}x} = \lim_{h\to 0} \frac{\cos(x+h) - \cos(x)}{h}
$$

As before, we are going to apply trigonometric identities to avoid the $1/h$ and then work limits separately. This time, the property is as follows $\cos(x+h) - \cos(x) = -2\sin\left(x + \frac{h}{2}\right)\sin\left(\frac{h}{2}\right)$.

$$
\begin{aligned}
\frac{\mathrm{d}\cos(x)}{\mathrm{d}x} &= \lim_{h\to 0} \frac{-2\sin\left(x + \frac{h}{2}\right)\sin\left(\frac{h}{2}\right)}{h} \\
&= \left[\lim_{h\to 0} -\sin\left(x + \frac{h}{2}\right)\right]\left[\lim_{h\to 0} \frac{2\sin\left(\frac{h}{2}\right)}{h}\right]
\end{aligned}
$$

Computing the limits separately

$$
\begin{aligned}
\lim_{h\to 0} -\sin\left(x + \frac{h}{2}\right) &= -\sin\left(x + \frac{0}{2}\right) \\
&= -\sin(x)
\end{aligned}
$$

As before, let $u = \frac{h}{2}, \quad h = 2u$ and as $h \to 0$, $u \to 0$

$$
\begin{aligned}
\lim_{h\to 0} \frac{2\sin\left(\frac{h}{2}\right)}{h} &= \lim_{u\to 0} \frac{2\sin(u)}{2u} \\
&= \cancelto{1}{\lim_{u\to 0} \frac{\sin(u)}{u}} \\
&= 1
\end{aligned}
$$

Replacing the results

$$
\begin{aligned}
\frac{\mathrm{d}\cos(x)}{\mathrm{d}x} &= \left[\lim_{h\to 0} -\sin\left(x + \frac{h}{2}\right)\right]\left[\lim_{h\to 0} \frac{2\sin\left(\frac{h}{2}\right)}{h}\right] \\
&= [-\sin(x)]\,[1] \\
&= -\sin(x)
\end{aligned}
$$